

- Lakretz, Yair, German Kruszewski, Theo Desbordes, Dieuwke Hupkes, Stanislas Dehaene, and Marco Baroni. 2019. The emergence of number and syntax units in LSTM language models. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, pages 11–20, Association for Computational Linguistics, Minneapolis, Minnesota.
- Lasri, Karim, Tiago Pimentel, Alessandro Lenci, Thierry Poibeau, and Ryan Cotterell. 2022. Probing for the usage of grammatical number. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 8818–8831, Association for Computational Linguistics, Dublin, Ireland.
- Lepori, Michael A., Thomas Serre, and Ellie Pavlick. 2024. Uncovering intermediate variables in transformers using circuit probing. In *First Conference on Language Modeling*.
- Lewis, David. 1986. Causation. In Tim Crane and Katalin Farkas, editors, *Philosophical Papers II*. Oxford University Press, pages 159–213.
- Lewis, David. 2000. Causation as influence. *The Journal of Philosophy*, 97(4):182–197.
- Lewis, David K. 1973. *Counterfactuals*. Blackwell, Malden, Massachusetts.
- Li, Jiwei, Will Monroe, and Dan Jurafsky. 2017. Understanding neural networks through representation erasure.
- Li, Kenneth, Aspen K Hopkins, David Bau, Fernanda Viégas, Hanspeter Pfister, and Martin Wattenberg. 2023. Emergent world representations: Exploring a sequence model trained on a synthetic task. In *The Eleventh International Conference on Learning Representations*.
- Li, Maximilian, Xander Davies, and Max Nadeau. 2024. Circuit breaking: Removing model behaviors with targeted ablation. *arXiv preprint arXiv:2309.05973*.
- Li, Victoria R., Jenny Kaufmann, Martin Wattenberg, David Alvarez-Melis, and Naomi Saphra. 2025. Can interpretation predict behavior on unseen data? *arXiv preprint 2507.06445*.
- Lipton, Zachary C. 2018. The mythos of model interpretability: In machine learning, the concept of interpretability is both important and slippery. *Queue*, 16(3):31–57.
- Liu, Nelson F., Matt Gardner, Yonatan Belinkov, Matthew E. Peters, and Noah A. Smith. 2019a. Linguistic knowledge and transferability of contextual representations. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, pages 1073–1094, Association for Computational Linguistics, Minneapolis, Minnesota.
- Liu, Yinhan, Mylène Ott, Naman Goyal, Jingfei Du, Mandar Joshi, Danqi Chen, Omer Levy, Mike Lewis, Luke Zettlemoyer, and Veselin Stoyanov. 2019b. RoBERTa: A robustly optimized BERT pretraining approach. *arXiv preprint arXiv:1907.11692*.
- Lundberg, Scott M. and Su-In Lee. 2017. A unified approach to interpreting model predictions. In *Proceedings of the 31st International Conference on Neural Information Processing Systems, NIPS'17*, page 4768–4777, Curran Associates Inc., Red Hook, NY, USA.
- Lyu, Qing, Marianna Apidianaki, and Chris Callison-Burch. 2024. Towards faithful model explanation in NLP: A survey. *Computational Linguistics*, 50(2):657–723.
- Ma, Weicheng, Kai Zhang, Renze Lou, Lili Wang, and Soroush Vosoughi. 2021. Contributions of transformer attention heads in multi- and cross-lingual tasks. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*, pages 1956–1966, Association for Computational Linguistics, Online.
- Madsen, Andreas, Siva Reddy, and Sarath Chandar. 2022. Post-hoc interpretability for neural NLP: A survey. *ACM Comput. Surv.*, 55(8).
- Makelov, Aleksandar, Georg Lange, and Neel Nanda. 2024. Towards principled evaluations of sparse autoencoders for interpretability and control. In *ICLR 2024 Workshop on Secure and Trustworthy Large Language Models*.
- Marks, Samuel, Can Rager, Eric J Michaud, Yonatan Belinkov, David Bau, and Aaron Mueller. 2025. Sparse feature circuits: Discovering and editing interpretable causal graphs in language models. In *The Thirteenth International Conference on Learning Representations*.
- Marks, Samuel and Max Tegmark. 2023. The geometry of truth: Emergent linear structure in large language model representations of true/false datasets. *arXiv preprint arXiv:2310.06824*.
- McClelland, James L. and David E. Rumelhart. 1985. Distributed memory and the representation of general and specific information. *Journal of experimental psychology. General*, 114 2:159–97.
- McDougall, Callum Stuart, Arthur Conmy, Cody Rushing, Thomas McGrath, and Neel Nanda. 2024. Copy suppression: Comprehensively understanding a motif in language model attention heads. In *Proceedings of the 7th BlackboxNLP Workshop: Analyzing and Interpreting*

- Neural Networks for NLP*, pages 337–363, Association for Computational Linguistics, Miami, Florida, US.
- McGrath, Thomas, Matthew Rahtz, Janos Kramar, Vladimir Mikulik, and Shane Legg. 2023. The hydra effect: Emergent self-repair in language model computations. *arXiv preprint arXiv:2307.15771*.
- Méloux, Maxime, Silviu Maniu, François Portet, and Maxime Peyrard. 2025. Everything, everywhere, all at once: Is mechanistic interpretability identifiable? In *The Thirteenth International Conference on Learning Representations*.
- Meng, Kevin, David Bau, Alex J Andonian, and Yonatan Belinkov. 2022. Locating and editing factual associations in GPT. In *Advances in Neural Information Processing Systems*.
- Meng, Kevin, Arnab Sen Sharma, Alex J Andonian, Yonatan Belinkov, and David Bau. 2023. Mass-editing memory in a transformer. In *The Eleventh International Conference on Learning Representations*.
- Merullo, Jack, Carsten Eickhoff, and Ellie Pavlick. 2024a. Circuit component reuse across tasks in transformer language models. In *The Twelfth International Conference on Learning Representations*.
- Merullo, Jack, Carsten Eickhoff, and Ellie Pavlick. 2024b. Talking heads: Understanding inter-layer communication in transformer language models. *arXiv preprint arXiv:2406.09519*.
- Michaud, Eric J, Ziming Liu, Uzay Girit, and Max Tegmark. 2023. The quantization model of neural scaling. In *Thirty-seventh Conference on Neural Information Processing Systems*.
- Mikolov, Tomas, Kai Chen, Greg Corrado, and Jeffrey Dean. 2013a. Efficient estimation of word representations in vector space. *arXiv preprint arXiv:1301.3781*.
- Mikolov, Tomas, Martin Karafiát, Lukáš Burget, Jan Černocký, and Sanjeev Khudanpur. 2010. Recurrent neural network based language model. In *Interspeech*, pages 1045–1048.
- Mikolov, Tomas, Ilya Sutskever, Kai Chen, Greg S Corrado, and Jeff Dean. 2013b. Distributed representations of words and phrases and their compositionality. In *Advances in Neural Information Processing Systems*, volume 26, Curran Associates, Inc.
- Miller, Joseph, Bilal Chughtai, and William Saunders. 2024. Transformer circuit evaluation metrics are not robust. In *First Conference on Language Modeling*.
- Mohebbi, Hosein, Jaap Jumelet, Michael Hanna, Afra Alishahi, and Willem Zuidema. 2024. Transformer-specific interpretability. In *Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics: Tutorial Abstracts*, pages 21–26, Association for Computational Linguistics, St. Julian's, Malta.
- Moraffah, Raha, Mansoor Karami, Ruocheng Guo, Adrienne Raglin, and Huan Liu. 2020. Causal interpretability for machine learning-problems, methods and evaluation. *ACM SIGKDD Explorations Newsletter*, 22(1):18–33.
- Morcos, Ari S., David G.T. Barrett, Neil C. Rabinowitz, and Matthew Botvinick. 2018. On the importance of single directions for generalization. In *International Conference on Learning Representations*.
- Mozer, Michael C. and Paul Smolensky. 1988. Skeletonization: A technique for trimming the fat from a network via relevance assessment. In *Advances in Neural Information Processing Systems*, volume 1, Morgan-Kaufmann.
- Mueller, Aaron. 2024. Missed causes and ambiguous effects: Counterfactuals pose challenges for interpreting neural networks. In *ICML 2024 Workshop on Mechanistic Interpretability*.
- Mueller, Aaron, Atticus Geiger, Sarah Wiegrefe, Dana Arad, Iván Arcuschin, Adam Belfki, Yik Siu Chan, Jaden Fiotto-Kaufman, Tal Haklay, Michael Hanna, Jing Huang, Rohan Gupta, Yaniv Nikankin, Hadas Orgad, Nikhil Prakash, Anja Reusch, Aruna Sankaranarayanan, Shun Shao, Alessandro Stolfo, Martin Tutek, Amir Zur, David Bau, and Yonatan Belinkov. 2025. Mib: A mechanistic interpretability benchmark. In *Proceedings of the 42nd International Conference on Machine Learning*, PMLR.
- Mueller, Aaron, Yu Xia, and Tal Linzen. 2022. Causal analysis of syntactic agreement neurons in multilingual language models. In *Proceedings of the 26th Conference on Computational Natural Language Learning (CoNLL)*, pages 95–109, Association for Computational Linguistics, Abu Dhabi, United Arab Emirates (Hybrid).
- Nanda, Neel and Joseph Bloom. 2022. TransformerLens.
<https://github.com/TransformerLensOrg/TransformerLens>.
- Nanda, Neel, Andrew Lee, and Martin Wattenberg. 2023. Emergent linear representations in world models of self-supervised sequence models. In *Proceedings of the 6th BlackboxNLP Workshop: Analyzing and Interpreting Neural Networks for NLP*, pages 16–30, Association for

- Computational Linguistics, Singapore.
- Neo, Clement, Shay B Cohen, and Fazl Barez. 2024. Interpreting context look-ups in transformers: Investigating attention-MLP interactions. *arXiv preprint arXiv:2402.15055*.
- Odense, Simon and Artur d'Ávila Garcez. 2020. Layerwise knowledge extraction from deep convolutional networks. *arXiv preprint arXiv:2003.09000*.
- Oikarinen, Tuomas, Subhro Das, Lam M. Nguyen, and Tsui-Wei Weng. 2023. Label-free concept bottleneck models. In *The Eleventh International Conference on Learning Representations*.
- Olsson, Catherine, Nelson Elhage, Neel Nanda, Nicholas Joseph, Nova DasSarma, Tom Henighan, Ben Mann, Amanda Askell, Yuntao Bai, Anna Chen, Tom Conerly, Dawn Drain, Deep Ganguli, Zac Hatfield-Dodds, Danny Hernandez, Scott Johnston, Andy Jones, Jackson Kernion, Liane Lovitt, Kamal Ndousse, Dario Amodei, Tom Brown, Jack Clark, Jared Kaplan, Sam McCandlish, and Chris Olah. 2022. In-context learning and induction heads. *Transformer Circuits Thread*.
- Panickssery, Nina, Nick Gabrieli, Julian Schulz, Meg Tong, Evan Hubinger, and Alexander Matt Turner. 2024. Steering Llama 2 via contrastive activation addition. *arXiv preprint arXiv:2312.06681*.
- Park, Kiho, Yo Joong Choe, and Victor Veitch. 2023. The linear representation hypothesis and the geometry of large language models. In *Causal Representation Learning Workshop at NeurIPS 2023*.
- Paulo, Gonçalo, Alex Mallen, Caden Juang, and Nora Belrose. 2024. Automatically interpreting millions of features in large language models. *arXiv preprint arXiv:2410.13928*.
- Pearl, Judea. 2000. *Causality: Models, Reasoning, and Inference*. Cambridge University Press.
- Pearl, Judea. 2001. Direct and indirect effects. In *Proceedings of the Seventeenth Conference on Uncertainty in Artificial Intelligence*, pages 411–420, Morgan Kaufmann.
- Pearl, Judea. 2009. *Causality*, 2 edition. Cambridge University Press.
- Pennington, Jeffrey, Richard Socher, and Christopher Manning. 2014. GloVe: Global vectors for word representation. In *Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 1532–1543, Association for Computational Linguistics, Doha, Qatar.
- Prakash, Nikhil, Tamar Rott Shaham, Tal Haklay, Yonatan Belinkov, and David Bau. 2024. Fine-tuning enhances existing mechanisms: A case study on entity tracking. In *The Twelfth International Conference on Learning Representations*.
- Qiu, Linlu, Liwei Jiang, Ximing Lu, Melanie Sclar, Valentina Pyatkin, Chandra Bhagavatula, Bailin Wang, Yoon Kim, Yejin Choi, Nouha Dziri, and Xiang Ren. 2024. Phenomenal yet puzzling: Testing inductive reasoning capabilities of language models with hypothesis refinement. In *The Twelfth International Conference on Learning Representations*.
- Radford, Alec, Jeff Wu, Rewon Child, David Luan, Dario Amodei, and Ilya Sutskever. 2019. Language models are unsupervised multitask learners.
- Rai, Daking, Yilun Zhou, Shi Feng, Abulhair Saparov, and Ziyu Yao. 2024. A practical review of mechanistic interpretability for transformer-based language models. *arXiv preprint arXiv:2407.02646*.
- Rajamanoharan, Senthoooran, Arthur Conmy, Lewis Smith, Tom Lieberum, Vikrant Varma, János Kramár, Rohin Shah, and Neel Nanda. 2024. Improving dictionary learning with gated sparse autoencoders.
- Ravfogel, Shauli, Yanai Elazar, Hila Gonen, Michael Twiton, and Yoav Goldberg. 2020. Null it out: Guarding protected attributes by iterative nullspace projection. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 7237–7256, Association for Computational Linguistics, Online.
- Ravfogel, Shauli, Grusha Prasad, Tal Linzen, and Yoav Goldberg. 2021. Counterfactual interventions reveal the causal effect of relative clause representations on agreement prediction. In *Proceedings of the 25th Conference on Computational Natural Language Learning*, pages 194–209, Association for Computational Linguistics, Online.
- Ravfogel, Shauli, Francisco Vargas, Yoav Goldberg, and Ryan Cotterell. 2022. Adversarial concept erasure in kernel space. In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, pages 6034–6055, Association for Computational Linguistics, Abu Dhabi, United Arab Emirates.
- Ravichander, Abhilasha, Yonatan Belinkov, and Eduard Hovy. 2021. Probing the probing paradigm: Does probing accuracy entail task relevance? In *Proceedings of the 16th Conference of the European Chapter of the Association for Computational Linguistics: Main Volume*, pages